

Reading and writing files

Module 4 - Lesson 1

Overview

Files persist data between runs. Python's `open()` plus the `with` statement makes reading and writing text and structured data safe and simple.

Key Concepts

- `open(path, mode)` supports `r` (read), `w` (write), `a` (append), and `b` (binary) modes.
- `with open(...)` as `f` guarantees the file closes even on error.
- `f.read()`, `f.readline()`, and iterating over `f` handle different sizes of data.
- Write with `f.write()` or `print(..., file=f)`; use `writelines()` for lists.
- `json.dump/json.load` convert Python data to and from JSON.
- Paths come from the `pathlib` module for robust, cross-platform handling.

Example

```
import json

with open("notes.txt", "w") as f:
    f.write("line one\n")

with open("config.json") as f:
    config = json.load(f)

with open("config.json", "w") as f:
    json.dump(config, f, indent=2)
```

Summary

The `with` statement and `open()` read and write files safely. JSON keeps structured data portable, and `pathlib` makes paths reliable across operating systems.

Practice Checklist

- Write a list of lines to a file and read them back.
- Load a JSON file, modify it, and save it back.
- Use `pathlib` to build a path and confirm it resolves correctly.